



COIN CHANGE PROBLEM 6

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Given a value V , you have an infinite supply of N different coin denominations. Find the minimum number of coins required to make exactly V cents.

You must also print the coins used in an optimal solution.

If it is not possible to make the value V , print -1 .

Input Format

The first line contains an integer T , the number of test cases. For each test case: The first line contains two integers V and N . V — the target value. N — the number of available coin denominations. The second line contains N integers $C[i]$, representing the coin denominations.

Constraints

$$1 \leq T \leq 100 \quad 1 \leq V \leq 10^6 \quad 1 \leq N \leq 10^6 \quad 1 \leq C[i] \leq 10^6$$

Output Format

For each test case:

Print the coins used in a minimum-coin solution, separated by spaces. In the next line, print the minimum number of coins. If it is impossible to make V , print -1 .

Sample Input 0

```
7 2
2 4
```

Sample Output 0

```
-1
```

Sample Input 1

```
11 4
1 5 6 9
```

Sample Output 1

```
2
```

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Language

Python 3



```
1 import sys
2 from collections import deque
3
```



```
4 def solve():
5     input_data = sys.stdin.read().split()
6     if not input_data:
7         return
8
9     iterator = iter(input_data)
10
11 while True:
12     try:
13         V_str = next(iterator)
14     except StopIteration:
15         break
16
17     V = int(V_str)
18
19     try:
20         N = int(next(iterator))
21     except StopIteration:
22         break
23
24     coins = []
25     for _ in range(N):
26         try:
27             coins.append(int(next(iterator)))
28         except StopIteration:
29             break
30
31     if V == 0:
32         print(0)
33         continue
34
35     unique_coins = sorted(list(set(c for c in coins if c <= V)))
```

Line: 69 Col: 1



Upload Code as File



Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✓ Sample Test case 1

Input (stdin)

```
1 | 7 2
2 | 2 4
```

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Your Output (stdout)

```
1 | -1
```

Expected Output

```
1 | -1
```

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